

CSE 107 Fall 2026

Estimation and Poisson models

Week 9

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Friday, November 27: no lecture.

Parameter, estimator, estimate

Assume $X_1, \dots, X_n$ are iid, with $\mathbb{E}X_i = \mu$ and $\text{Var}(X_i) = \sigma^2$, where $0 < \sigma^2 < \infty$.

Object	Notation	Before observing the sample
Population mean	μ	Fixed; unknown
Sample measurements	$X_1, \dots, X_n$	Random variables
Sample-mean estimator	$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$	Random variable
Observed estimate	$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	A number after observation

The estimator is a rule applied to a random sample. The estimate is its observed value.

Mean and variance of the estimator

Use linearity of expectation and the variance of a sum of independent variables.

$$\mathbb{E}\bar{X}_n = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

An estimator T of μ is unbiased when $\mathbb{E}T = \mu$.

Standard error

The **standard error** of an estimator is its sampling standard deviation:

$$\text{SE}(T) = \sqrt{\text{Var}(T)}, \quad \text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Estimator of μ	Expectation	Variance	Standard error
X_1	μ	σ^2	σ
$\bar{X}_n$	μ	σ^2/n	$\sigma/\sqrt{n}$

The sampling distribution

Normal-sum theorem (stated). A linear combination of independent normal random variables is normal.

Normal observations

$$X_i \sim N(\mu, \sigma^2)$$

$$\bar{X}_n \sim N(\mu, \sigma^2/n) \text{ for every } n$$

General iid observations

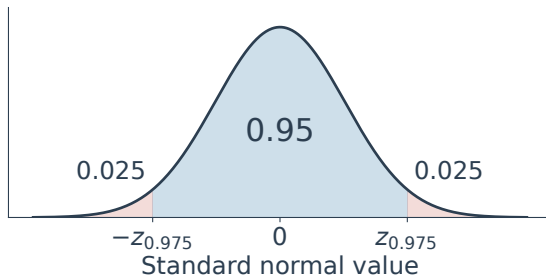
$$\mathbb{E}X_i = \mu, \text{Var}(X_i) = \sigma^2 \in (0, \infty)$$

The CLT gives an approximately normal standardized mean for suitably large n

A central normal probability

Write $z_p = \Phi^{-1}(p)$. Thus $z_{0.975} \approx 1.96$ and

$$P(-z_{0.975} \leq Z \leq z_{0.975}) = 0.95, \quad Z \sim N(0, 1).$$



For confidence level $1 - \alpha$, replace $z_{0.975}$ by $z_{1-\alpha/2}$, with $0 < \alpha < 1$.

Deriving the confidence interval

Assume iid normal observations and **known** $\sigma > 0$. Put $s = \sigma/\sqrt{n}$ and $z = z_{1-\alpha/2}$.

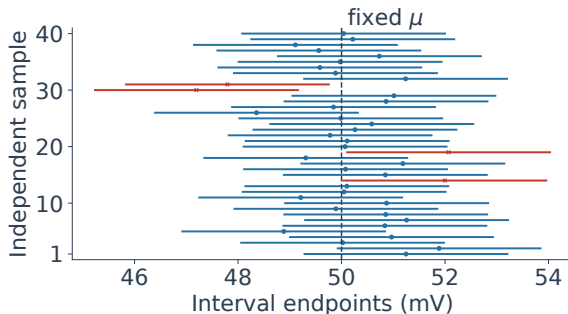
$$P_{\mu} \left(-z \leq \frac{\bar{X}_n - \mu}{s} \leq z \right) = 1 - \alpha.$$

Solve the event inequalities for the fixed parameter μ .

The same construction has approximate coverage when normality is justified only by the CLT.

Repeated-sampling coverage

Each row uses a fresh sample: $n = 100$, $X_i \sim N(\mu, 10^2)$, with fixed $\mu = 50$.



36 of these 40 intervals cover μ ; red intervals miss it. A finite run need not give exactly 95%. Simulated illustration; seed 20261123. Coverage describes the procedure over repeated samples.

Sensor measurements

A sensor gives iid $N(\mu, 10^2)$ measurements in mV; $\sigma = 10$ mV is known. For $n = 100$, the observed mean is $\bar{x} = 52$ mV. Find a 95% confidence interval for μ and interpret its coverage.

First identify the model and the standard error; then substitute the observed mean.

Planning interval width

With known σ , the half-width is $h_n = z_{1-\alpha/2}\sigma/\sqrt{n}$. How large must n be to make $h_n \leq h$, where $h > 0$?

The width algebra is exact. Coverage is exact under normality, approximate under a CLT justification.

An interval calculation

Independent normal measurements have unknown mean μ and known standard deviation 6 units. A sample of 144 gives $\bar{x} = 20$.

- 1 Find the standard error of the sample mean.
- 2 Construct the 95% confidence interval using $z_{0.975} \approx 1.96$.

Conditions for coverage

Check	Consequence
An iid sample	$\text{Var}(\bar{X}_n) = \sigma^2/n$
Known $0 < \sigma^2 < \infty$	The stated interval uses the true standard error
Normal population	Coverage $1 - \alpha$ for every n
Only a CLT approximation	Approximate coverage; check the model and sample size
Observed endpoints	Fixed after sampling; the procedure has the coverage property

Next: model a nonnegative integer count and derive its mean and variance.

Counts in a fixed interval

Let K be the number of customers arriving during a specified ten-minute observation window.

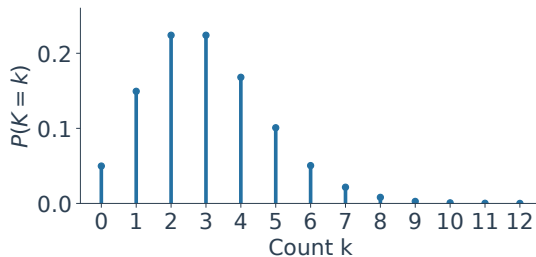
$$K \in \{0, 1, 2, \dots\}.$$

Binomial count	Poisson count
Number of successes in a fixed number n of trials	A model for a count with no fixed upper bound
Parameters n, p	Parameter m , the mean count in the window

Poisson mass function

For $m > 0$, write $K \sim \text{Poisson}(m)$ when

$$p_K(k) = e^{-m} \frac{m^k}{k!}, \quad k = 0, 1, 2, \dots; \quad p_K(k) = 0 \text{ otherwise.}$$



Example: $m = 3$. Stems stop at 12; the omitted upper-tail probability is below 0.000017. For $m = 0$, the law is concentrated at 0.

Masses sum to one

Use the exponential-series identity $e^x = \sum_{k=0}^{\infty} x^k/k!$. Show that the proposed masses are nonnegative and sum to one.

The exponential series is the only analytic identity needed here.

Poisson mean

Let $K \sim \text{Poisson}(m)$, with $m > 0$. Start from the definition:

$$\mathbb{E}K = \sum_{k=0}^{\infty} k e^{-m} \frac{m^k}{k!}.$$

The parameter m is the mean count in this particular observation window.

Poisson variance

Compute $\mathbb{E}[K(K - 1)]$ first; then use $K^2 = K(K - 1) + K$.

For a Poisson law, the mean and variance have the same numerical value.

Coffee-shop arrivals

Model the number K of arrivals in one specified ten-minute window as $\text{Poisson}(3)$.

- ① Find $P(K = 2)$.
- ② Find $P(K \geq 1)$.

This single-window law alone does not specify counts in other windows.

Rate and interval length

In a homogeneous Poisson process, an interval of duration Δ has mean count

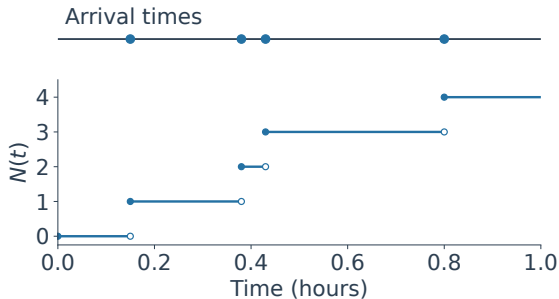
$$m = \lambda\Delta.$$

Rate λ	Duration	In hours	Mean $m = \lambda\Delta$
18 per hour	10 minutes	1/6	3
18 per hour	1 hour	1	18
4 per hour	15 minutes	1/4	1

Changing the time unit changes the numerical rate, not the predicted count.

A counting process

$N(t)$ counts arrivals during $(0, t]$, with $N(0) = 0$.



Paths are integer valued and nondecreasing. The right-continuous convention includes an arrival at the right endpoint: $(a, b]$ contains $N(b) - N(a)$ arrivals.

Independent stationary increments

A **homogeneous Poisson process** with rate $\lambda > 0$ has $N(0) = 0$ and:

① For $0 \leq a < b$,

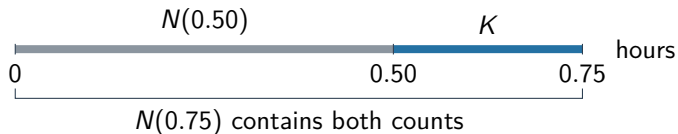
$$N(b) - N(a) \sim \text{Poisson}(\lambda(b - a)).$$

② Counts in any finite collection of disjoint time intervals are jointly independent.

Check the intervals before making an independence claim.

A later observation interval

Arrivals form a homogeneous Poisson process at rate $\lambda = 4$ per hour. Let K count arrivals between 30 and 45 minutes after observation begins.



Find the law of K and $P(K \leq 1)$.

Model and units check

Before calculating a count probability:

- ① State the Poisson law or the process assumptions.
 - ② Express λ and the interval duration in compatible units.
 - ③ Use $m = \lambda\Delta$ in the mass function.
 - ④ Claim independence only for counts on disjoint intervals under the process model.
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November 30: combine counts on disjoint intervals and compare overlapping counts.

A guaranteed interval from Chebyshev

For iid observations with known variance $\sigma^2 > 0$, Chebyshev gives

$$P\left(|\bar{X}_n - \mu| \geq \frac{\sigma}{\sqrt{n\alpha}}\right) \leq \alpha.$$

A finite-sample lower bound on coverage is different from exact equality or a CLT approximation.

A rare-success binomial limit

Let $K_n \sim \text{Binomial}(n, m/n)$, with fixed $m > 0$. For fixed integer $k \geq 0$ and sufficiently large n ,

$$P(K_n = k) = \frac{m^k}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) \left(1 - \frac{m}{n}\right)^n \left(1 - \frac{m}{n}\right)^{-k}.$$

This is a different limit from the fixed- p binomial CLT. The product is 1 when $k = 0$.

Estimating the standard error

If σ is unknown, define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad \widehat{SE}(\bar{X}_n) = \frac{S}{\sqrt{n}}, \quad n \geq 2.$$

Unknown-variance small-sample methods are outside this week's required material.

Estimating a Poisson rate

Observe counts $X_1, \dots, X_n$ in disjoint windows, each of length $\Delta > 0$, from a homogeneous Poisson process of rate λ .

$$\hat{\lambda} = \frac{\overline{X}_n}{\Delta}.$$

With few counts, a plug-in normal approximation can be poor; zero estimated SE does not establish certainty.

Additional count practice

Suppose arrivals form a homogeneous Poisson process at rate 6 per hour.

- ① Find the chance of exactly two arrivals during a specified 20-minute interval.
- ② Give the mean and variance of that count.
- ③ Would a constant average rate alone justify the same probability?

Selected reading

Harchol-Balter, *Introduction to Probability for Computing* (supplied 2024 version):

- pp. 255–257: estimators and the sample mean; pp. 334–336 through (19.8): known-variance intervals.
- p. 49: Poisson masses; pp. 216–217, Definitions 12.6–12.10: processes and rate.

DeGroot and Schervish, *Probability and Statistics*, 4th ed.:

- pp. 311–312: normal sample mean and known-variance interval; p. 465 (Definition 8.1.1); p. 507 (unbiasedness).
- pp. 486–487: confidence definition and repeated-sample meaning only.
- pp. 288–290 through Theorem 5.4.2: Poisson moments; p. 293, Definition 5.4.2: process increments.

Printed page numbers. These are selected passages, not whole-chapter assignments. The course notes are self-contained.